

AAYUSH DESAI

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SUMMARY

Backend Software Engineer with 5 years in healthcare technology, specializing in scalable distributed systems, EHR integrations, high-throughput APIs, and event-driven architecture in C#/.NET. Built and deployed production multi-agent systems and RAG pipelines as personal projects alongside professional work.

SKILLS

- **Languages & Frameworks:** C#, ASP.NET Core (5–10), Python, TypeScript, SQL, Ruby on Rails, Entity Framework Core
- **AI & Agent Systems:** Semantic Kernel, RAG, Qdrant, RAGAS, Langfuse, MCP, Ollama, Groq, Voyage AI, Azure OpenAI
- **Cloud & Infrastructure:** Azure (App Services, Service Bus, Event Hub), AWS (EC2, S3, SQS, Lambda), Docker, Kubernetes, Redis, GitHub Actions
- **Backend Engineering:** Microservices, Event-Driven Design, REST APIs, Circuit Breakers, Caching, Concurrency, HIPAA Compliance

PROJECTS

ChefAgent — Multi-Agent Cooking Assistant

chefagent.vercel.app ◇ [GitHub](#)

- Built 4-agent system (Orchestrator, RecipeAgent, DietAgent, PlannerAgent) via **Semantic Kernel** over a **RAG pipeline** of 52k recipes (Qdrant + Voyage AI embeddings) — improved context relevance from 0.470 to 0.578 through spell correction and corpus expansion
- Designed provider-agnostic LLM and embedding abstractions enabling **3 production provider swaps** without touching the agent layer — reduced query latency from ~14,000ms to ~651ms
- Deployed with 5-layer guardrails, Langfuse observability (14 span types), and custom RAGAS eval pipeline with an **87%** pass rate across 60 hand-curated test cases, deployed on Railway + Vercel

mcp-dotnet-diagnostics — .NET Runtime Diagnostics MCP Server

[GitHub](#) ◇ [NuGet](#)

- Open-source **MCP server** (NuGet global tool) exposing 7 .NET runtime diagnostic tools (memory, GC, threads, EventCounters) — giving AI assistants direct access to diagnose LOH fragmentation and thread starvation in live .NET processes; 34 tests, CI/CD auto-publish, listed on Glama and mcpservers.org

EXPERIENCE

Software Engineer — Carenet Health

Dallas, TX — Jul 2021 – Mar 2026

- Led **IQ-Prompt Integration** — a .NET API gateway with API key auth, request validation, audit logging, and allowed-action enforcement, enabling LLM-based agents to invoke backend services across AWS and Azure via Model Context Protocol (MCP)
- Trained 3 junior analysts on system workflows and production practices; SME and technical lead across MCP integration, Ruby-to-.NET migration, and Patient Journey initiatives
- Led modernization of legacy Ruby systems into **.NET 6–8 microservices** via feature flags, with backward-compatible APIs, zero-downtime migration, and unit and integration test coverage. **Impact:** Production incidents –40%
- Designed real-time scheduling and demographic synchronization across Epic, Athena, Meditech using REST + HL7 transformations. **Impact:** API latency –25%
- Built a concurrency-safe **availability & slot computation engine** using parallel execution and Redis and in-memory caching. **Impact:** 40% faster slot searches at scale
- Led backend for the **Patient Intake Portal**, collaborating with architects to deliver OCR document capture, Stripe payments, and AES-encrypted data handling for self-service capabilities
- Architected a HIPAA-compliant **event streaming pipeline** processing 1K–10K booking events daily with durable retries and idempotency, publishing to Azure Event Hub
- Drove consolidation of 5+ services into a shared orchestration layer (.NET 8, Hangfire, Azure Service Bus, Event Hub), reducing cross-team integration overhead
- Independently stood up the AdvanceMD EHR connector end-to-end within a month, enabling a new enterprise client segment

EDUCATION

Master of Computer Science — Syracuse University

2019 – 2021

Bachelor of Computer Engineering — Gujarat Technological University

2015 – 2019

CERTIFICATIONS

Building Intelligent Agent Workflows — Microsoft / Coursera

Jan 2026

AI Agents: From Prompts to Multi-Agent Systems — UC Davis / Coursera

Jan 2026

AI Agent Fundamentals with Azure AI Foundry — Microsoft / Coursera

Feb 2026